



MAE 3050 - Galider the Good

Watch us defy gravity

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Introduction

The goal of this project was to design a glider with a minimized cruise velocity while ensuring the aircraft can still travel the required distance of 20 meters when launched from a height of 5 meters. In order to achieve this goal, it was necessary to design a glider such that it would fly as close to stall as possible, maximize Cl/Cd such that it flies as efficiently as possible, maximize Cl such that the plane will fly as slow as possible without stalling, and maximize static margin to increase stability. In order to make design decisions, it was assumed that the velocity would be constant and the glider would trim to fly in a steady state. An iterative design process was used, including analytical calculations and XFLR5 analyses, to hone in on the finalized design. Initial estimates were made using general knowledge, best practices, and concepts from MAE 3050 lecture and discussion. Additionally, several test flights of the first iteration were conducted, and the results from them were used to inform the final design. The final design is a glider with a large aspect ratio, a relatively large static margin, and a maximized wingspan designed to fly at a low cruise velocity and high angle of attack.

Glider Description

The initial strategy behind designing the glider was to maximize the wingspan and the coefficient of lift in order to fly as close to stall as possible, covering the required distance of 20 meters in the slowest possible time. Additionally, the static margin was maximized within a reasonable range in order to make the glider as stable as possible and resist perturbations. Given how lightweight the airframe is, it was critical to do this. Several assumptions were made during the design process to keep the project within a reasonable scope. The main assumption was that the glider would fly in steady state. Not only does this assumption keep the analysis within the scope of the course, but it is also reasonable to make for most of the flight, given that the glider flies without any external inputs or disturbances in cruise. Following on from this assumption, it was also assumed that the glider would fly at a constant velocity and at a constant angle of attack. Using a steady state analysis, these two assumptions are also reasonable in the context of an undisturbed glider flight. Lastly, the process of designing a glider or aircraft is not a closed problem and does not have a single solution. This means that several decisions were necessarily made based on best practices, outside knowledge, and other resources, not strict calculations. That process is documented here and was eventually validated using analysis software such as XFLR5 and initial flight tests.

The team began by setting a goal angle of attack of 8 degrees and a target cruise velocity of 3 m/s. To find the Reynolds number that the glider will be estimated to be flying at, a range of reasonable chord lengths was used, producing a Reynolds number range of 35,000 to 45,000. This range was used for analysis until a final chord length was determined. Additionally, a mass

estimate of 112 grams was used in the initial calculations until the CAD was made, and the estimate could be refined.

The design process began with the selection of an airfoil. Several targets were considered in the selection process, including having a large CL/CD at the target alpha, having a large CL at the target alpha, having a relatively low C_m at the target alpha, and having gentle stall characteristics. An analysis of multiple airfoil candidates was conducted using XFLR5 over the relevant Reynolds number range, and the S3021 was chosen because it performed best across all target categories throughout the range. The full analysis results are shown below. This analysis also produced an estimate of the CL at the target angle of attack, which was used to refine the calculations further.

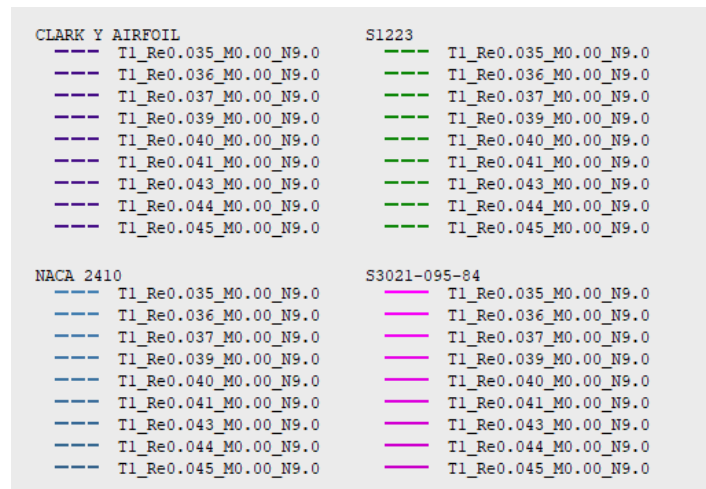


Figure 1: Airfoils Analyzed

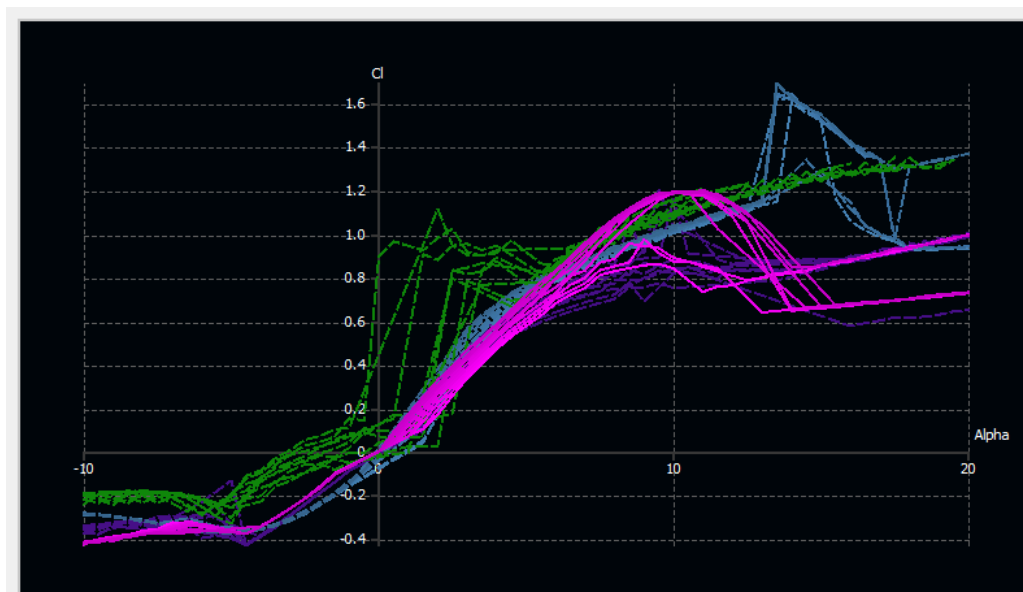


Figure 2: CL vs Alpha for Analyzed Airfoils

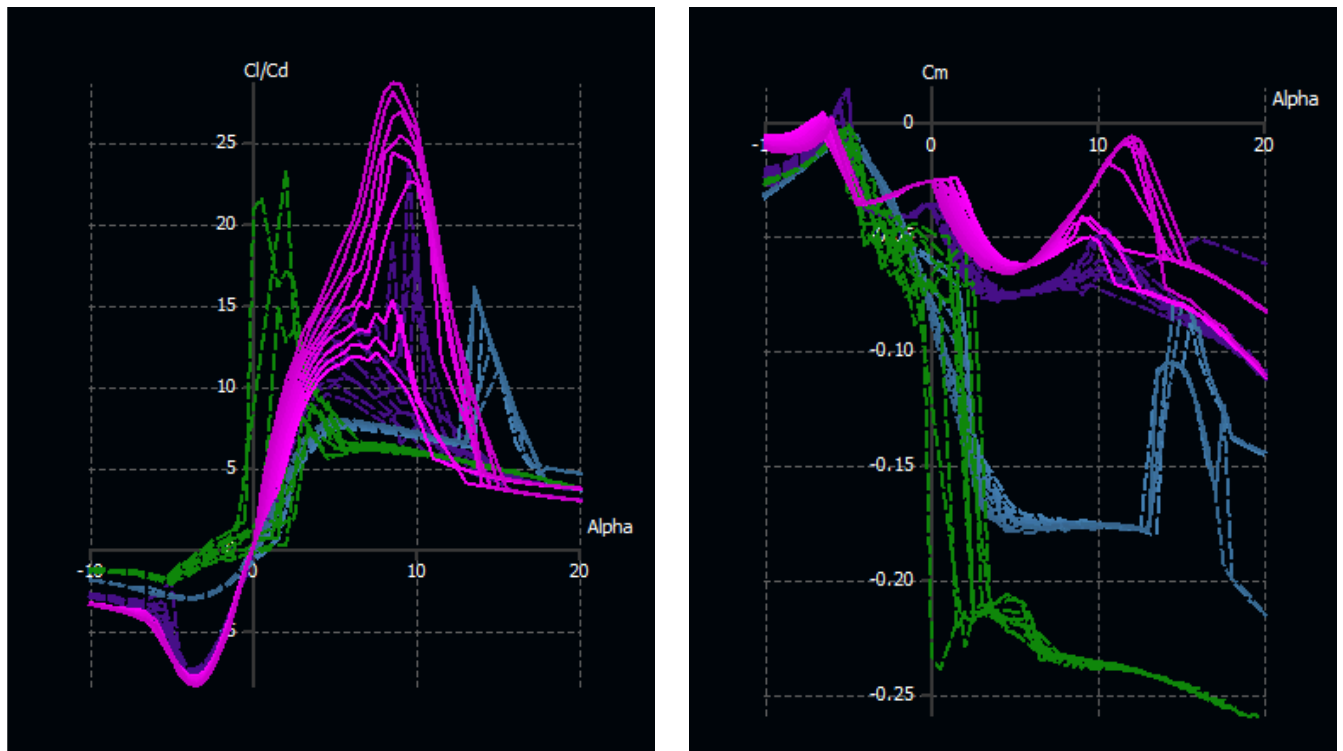


Figure 3: CL/CD vs α and C_m vs α for Analyzed Airfoils

As mentioned, the airfoil analysis conducted in XFLR5 showed that the S3021, the hot pink lines, performed the best according to the set targets. Most notably, it exhibited a CL/CD peak at the target α , a relatively high CL at the target α , and gentle stall characteristics beyond that. This airfoil was used for both the initial and final designs of the glider.

Given the material constraints, an initial wingspan of 1.5 m was chosen as the maximum feasible. However, the first round of construction showed that it was not possible to build a stiff 1.5 m wing, but it was possible to build a 1.4 m wing. The reason for this is that using the remaining 10 cm of spar as a doubler close to the root of the wing provided the extra stiffness necessary. Therefore, the final design has a wingspan of 1.4 m. Once the wingspan was chosen, the lift equation was used to calculate the chord length of 0.183 m, and then the chord length was used to refine the Reynolds number. Once the Reynolds number was determined, the airfoil analysis was inspected again to ensure that the initial selection, using the range, was still valid. The initial design and the final design used the same methodology in determining the chord, but the differing wingspan caused differing chord lengths.

Once the wing dimensions were settled on, the empennage dimensions and fuselage length were calculated. Once again, material constraints were the driving factor behind one of the key dimensions. The fuselage length was set to be 0.91 m, the length of one of the given spruce sticks. From there, an estimate of the wing placement was made based on research into other gliders and general best practices outlined in Daniel Raymer's textbook on conceptual aircraft design. Based on these considerations, the leading edge of the wing was placed approximately 35% back from the nose of the aircraft, and the tail arm distance could then be set by subtracting that distance and the chord length from the total fuselage length. Using the tail volume coefficient method, the sizes of the horizontal and vertical stabilizers were calculated. These values changed between the initial and final iterations of the glider because of the changes made to the wingspan and a desire to increase the static margin of the aircraft. Once all the dimensions were determined, a CAD was made to validate the initial mass estimates.

Before finalizing the design and in order to determine the necessary ballast weight, a detailed and thorough stability analysis was conducted using XFLR5. CAD mass estimates, along with wing and empennage dimensions, were used to construct the plane in the software. Additionally, the horizontal and vertical stabilizers' airfoil was set to NACA 0004 even though the true design will use a simple flat plate. XFLR5 cannot model flat plates, so this was a required approximation. The stability analysis showed that the design should function optimally for the competition. The results of the analysis are tabulated in the following section, but in brief, XFLR5 showed that the CL and CL/CD for the aircraft were close to initial estimates of 0.8 and 1.14. Additionally, the trimmed alpha and cruise speed were optimal. Most importantly, the stability derivatives of the aircraft were all in the stable regime except for the spiral mode. For a stable system, the eigenvalues of each of the modes must be negative, meaning that the oscillations will decay to zero. This was the case for all of the longitudinal and lateral modes of the aircraft except for the spiral mode. However, from research and past project team experience, it is expected that the spiral mode will be unstable. In unstable mode, the goal is to maximize the time to double, which in this case was just over 1 second. The results of the stability analysis are shown and discussed below.

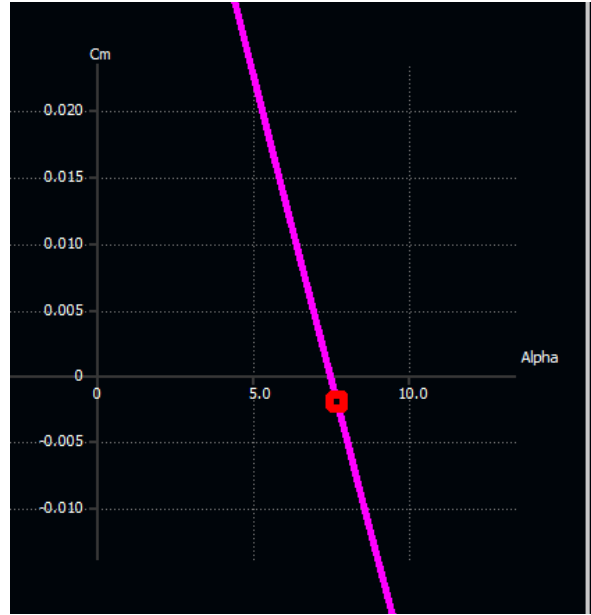


Figure 4: Cm vs Alpha from Stability Analysis

The first and most critical result of the stability analysis comes from the Cm vs Alpha graph, with the red dot indicating the trim position of the aircraft (where $C_m = 0$). XFLR5 indicates that the glider will trim to an angle of attack of 7.69 degrees, the result used in the team's calculations. Additionally, the slope of the line, representing the static margin, is quite steep, indicating a relatively large static margin. Calculating the static margin using the locations of the aerodynamic center and center of gravity of the aircraft gave a result of -10.3% of the maximum aerodynamic chord (MAC). Empirical evidence indicates that the magnitude of the static margin should be between 5% and 15% of MAC, so this result was satisfactory.

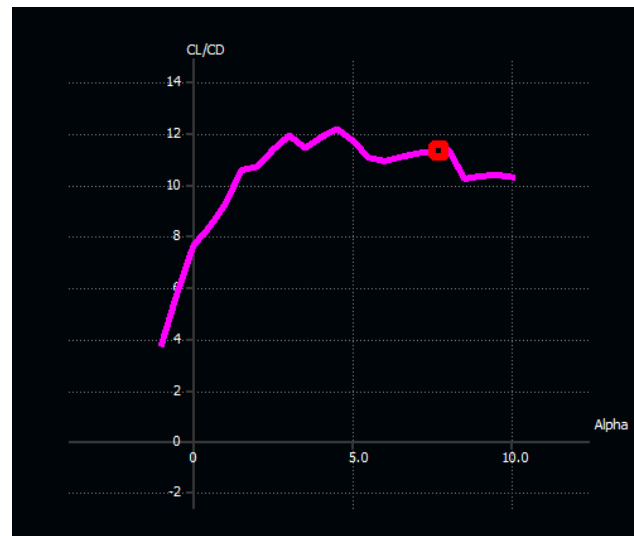
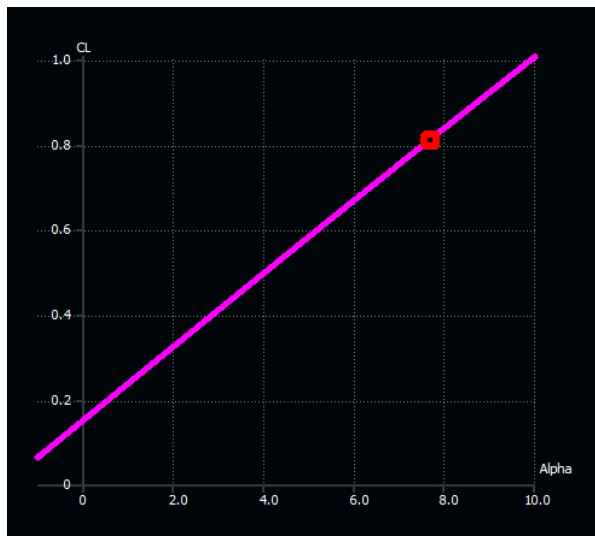


Figure 5: CL vs Alpha and CL/CD vs Alpha from Stability Analysis

Additional results from the stability analysis are that the CL at the trim condition will be 0.81172 and that CL/CD at the trim condition will be 11.3. The CL result is consistent with the initial airfoil analysis and validates the analytical calculations done. The CL/CD result indicates that with a starting height of 5 m, the glider should have more than enough range to complete the required distance. Calculations based on this result are shown in the calculations section below. Finally, the cruise speed at the trim condition was found to be 2.938 m/s. When performing the final calculations, these values were used instead of the initial estimates to show the characteristics, performance, and geometry of the plane as it will be flown.

Calculations

Chord length using a maximum wingspan of 1.4 meters

$$\text{Mass estimate} = 112 \text{ g}$$

$$L = \text{Mass} \cdot 9.806 \text{ m/s}^2$$

$$L = 0.112 \text{ kg} \cdot 9.806 \text{ m/s}^2$$

$$L = 1.0995$$

$$C_L = 0.81172$$

$$\rho = 1.225 \text{ kg/m}^3$$

$$V = 2.938 \text{ m/s}$$

$$b = 1.4 \text{ m}$$

$$L = \frac{1}{2} \rho V^2 C_L A$$

$$L = \frac{1}{2} \rho V^2 C_L (c \cdot b)$$

$$c = \frac{2L}{\rho V^2 C_L b}$$

$$c = \frac{2(1.0995)}{(1.225 \text{ kg/m}^3)(2.938 \text{ m/s})^2(0.81)(1.4)}$$

$$c = 0.143 \text{ m}$$

Aspect ratio

$$AR = \frac{b^2}{S} = \frac{b^2}{b \cdot c} = \frac{b}{c} = \frac{1.4 \text{ m}}{0.143 \text{ m}} = 7.65$$

Tail Coefficient

$$\text{Tail arm} = 0.50175$$

$$C_{HT} = \frac{L_{HT} S_{HT}}{C_W S_W}$$

$$\frac{(0.50175)(0.0308)}{(0.143)(0.2562)} = 0.33 = C_{HT}$$

$$S_{HT} = 0.44 \times 0.07$$

$$= 0.0308 \text{ m}^2$$

$$S_{VT} = 0.15 \times 0.07$$

$$= 0.0105 \text{ m}^2$$

$$C_{VT} = \frac{L_{VT} S_{VT}}{b_W S_W}$$

$$\frac{(0.50175)(0.0105)}{(1.4)(0.2562)} = 0.015 = C_{VT}$$

Flight speed

$$L = m \cdot g$$

$$L = (0.113)(9.806)$$

$$L = 1.1$$

$$L = \frac{1}{2} C_L \rho V^2 b c$$

$$V^2 = \frac{1.1}{\frac{1}{2}(0.81)(1.225)(0.183)(1.4)} = \sqrt{8.65} = V = 2.94 \text{ m/s}$$

Glide ratio

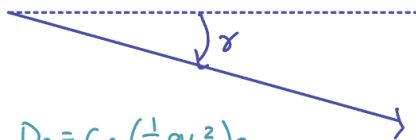
glide Ratio:

$$\tan(2.578) = \frac{5}{x}$$

$$\frac{111}{5} = 22.2$$

$$x = 111 \text{ m}$$

Sink rate



Define γ as glide path $\angle$

$$\begin{aligned} h &= 5 \text{ m} & mg\gamma &= T-D \\ x &= 20 \text{ m} & T &= 0 \end{aligned}$$

$$D_f = C_f \left(\frac{1}{2} \rho V_a^2 \right) S_{wet}$$

$$D_i = C_{Di} \left(\frac{1}{2} \rho u^2 S \right)$$

$$D_f = 0.007 \left(\frac{1}{2} (1.225) (2.938)^2 \right) (0.2562) = 0.0095 \text{ N}$$

$$D_i = 0.029 \left(\frac{1}{2} (1.225) (2.938)^2 \right) (0.2562) = 0.04 \text{ N}$$

$$D_{tot} = D_f + D_i = 0.05 \text{ N}$$

$$\gamma = \frac{-0.05 \text{ N}}{(0.113)(9.806)} = -0.045 \text{ rad/s} = -2.578$$



$$\cos(2.578) = \frac{20}{\text{path}} \quad \text{path} = 20.02 \text{ m}$$

$$20.02 \text{ m} = vt = 2.983 t \Rightarrow t = 6.71 \text{ s}$$

$$\text{falls } 5 \text{ m in } 6.71 \text{ s so } \frac{5}{6.71} = -0.745 \text{ m/s} = \text{sinkrate}$$

Pull back distance

$$\frac{1}{2} k x^2 = \frac{1}{2} m v^2$$

$$\frac{1}{2} (61) x^2 = \frac{1}{2} (0.095) (2.938)^2$$

$$30.5 x^2 = 0.41$$

$$x = 0.116 \text{ m}$$

Reynolds number,

$$Re = \frac{\rho V L}{\mu} = \frac{(1.225)(3)(0.183)}{1.81e^{-5}} = 37156.08$$

Mach number

$$M = \frac{V}{\sqrt{\gamma R T}} = \frac{2.938}{\sqrt{(1.4)(287)(288.15)}} = 0.0086$$

Technical Summary

Design Parameter	Value
Aspect Ratio	7.65
Planform Area	0.2562 m ²
Chord Length	0.183 m
Tail coefficient - C_{HT}	0.33
Tail coefficients - C_{VT}	0.015
CL	0.81172
CD	0.07178
CDi	0.029
CDf	0.007
Angle of attack	7.69 degrees
Flight time	6.71 secs
Flight Speed	2.938 m/s
Glide ratio	22.2
Sinkrate	-0.745 m/s
Reynolds Number	37156.08
Mach Number	0.0086
Pull-back Distance (k = 61 N/m)	0.116 m
Weight	113 g
Stall Speed	2.426 m/s

Design + Bill of Materials

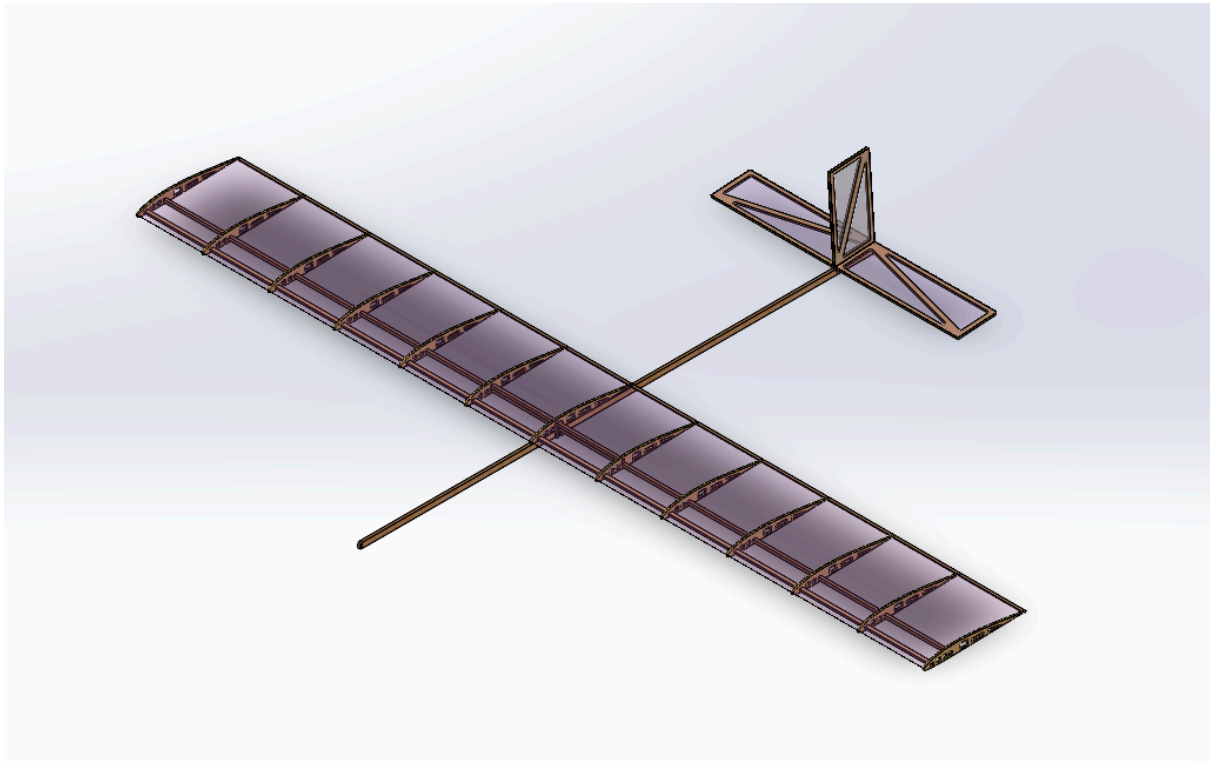


Figure 6: CAD Isometric View

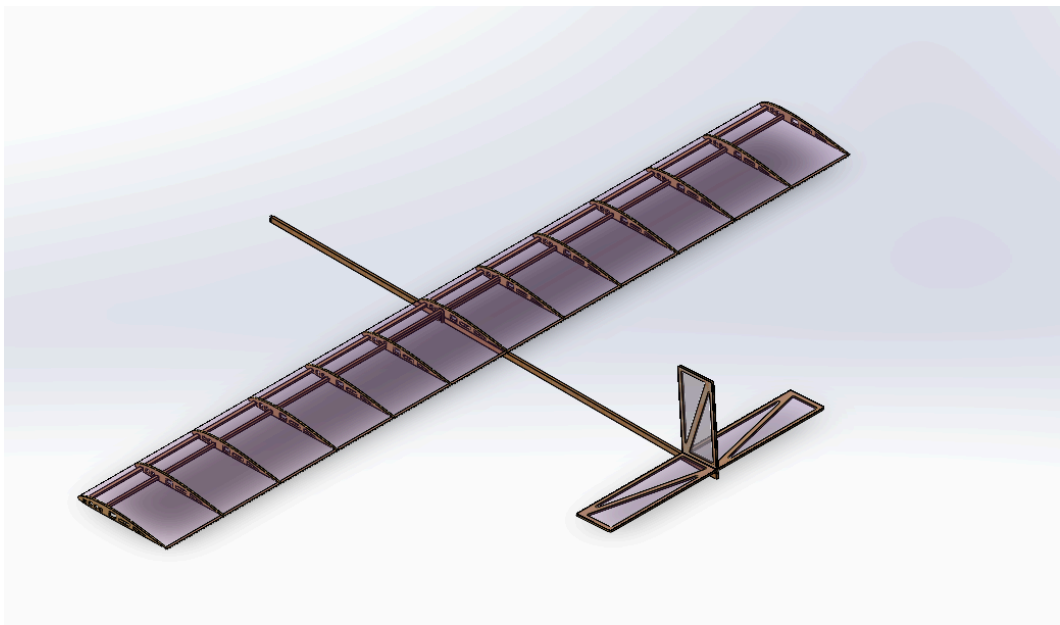


Figure 7: Rear Isometric View

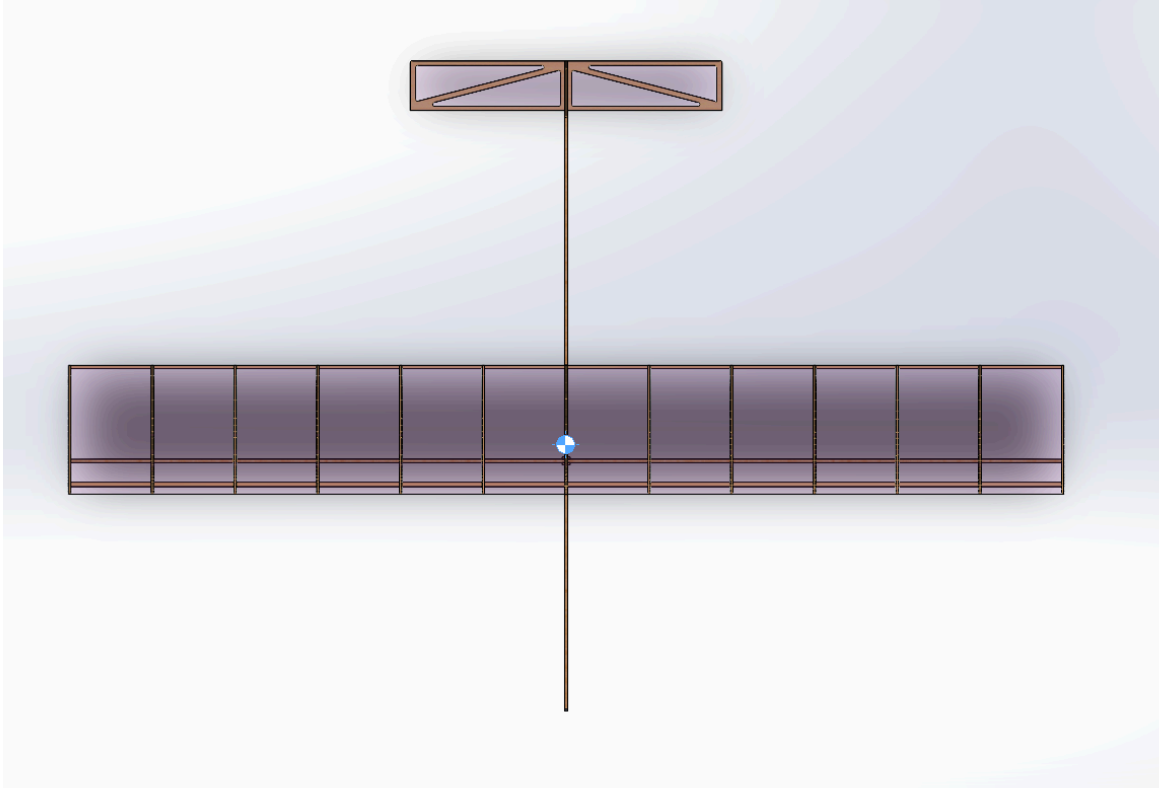


Figure 8: Top Down View with Center of Mass Indicated

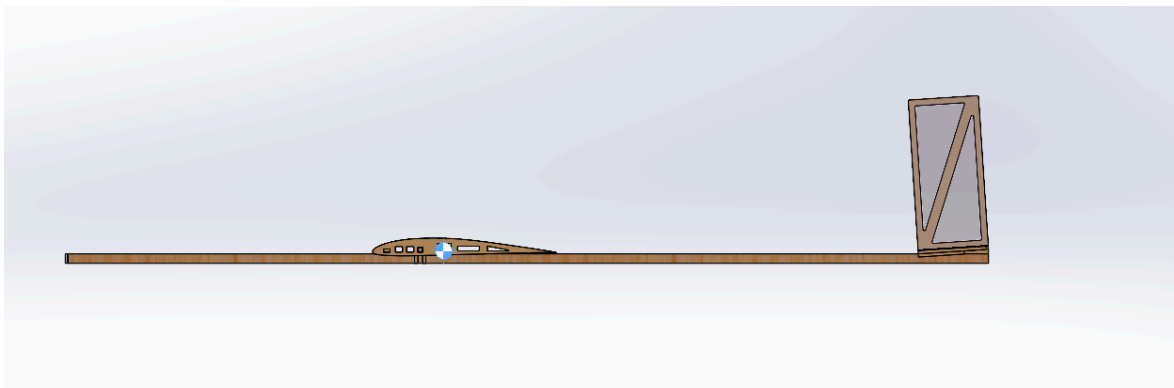


Figure 9: Side View with Center of Mass Indicated

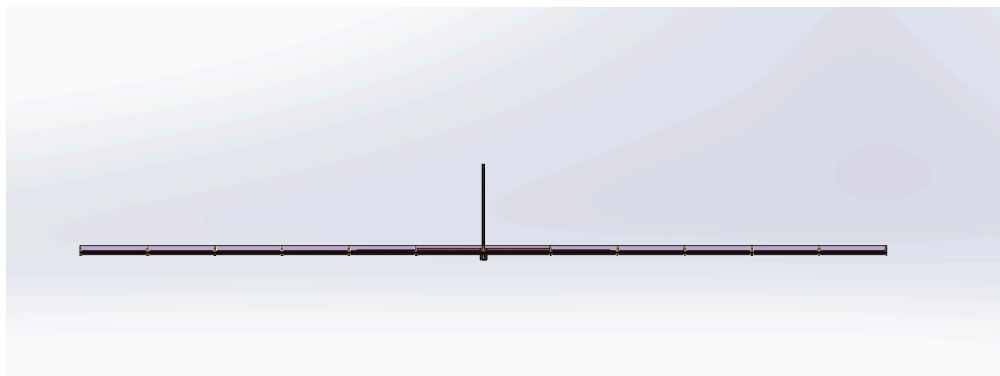


Figure 10: Frontal View

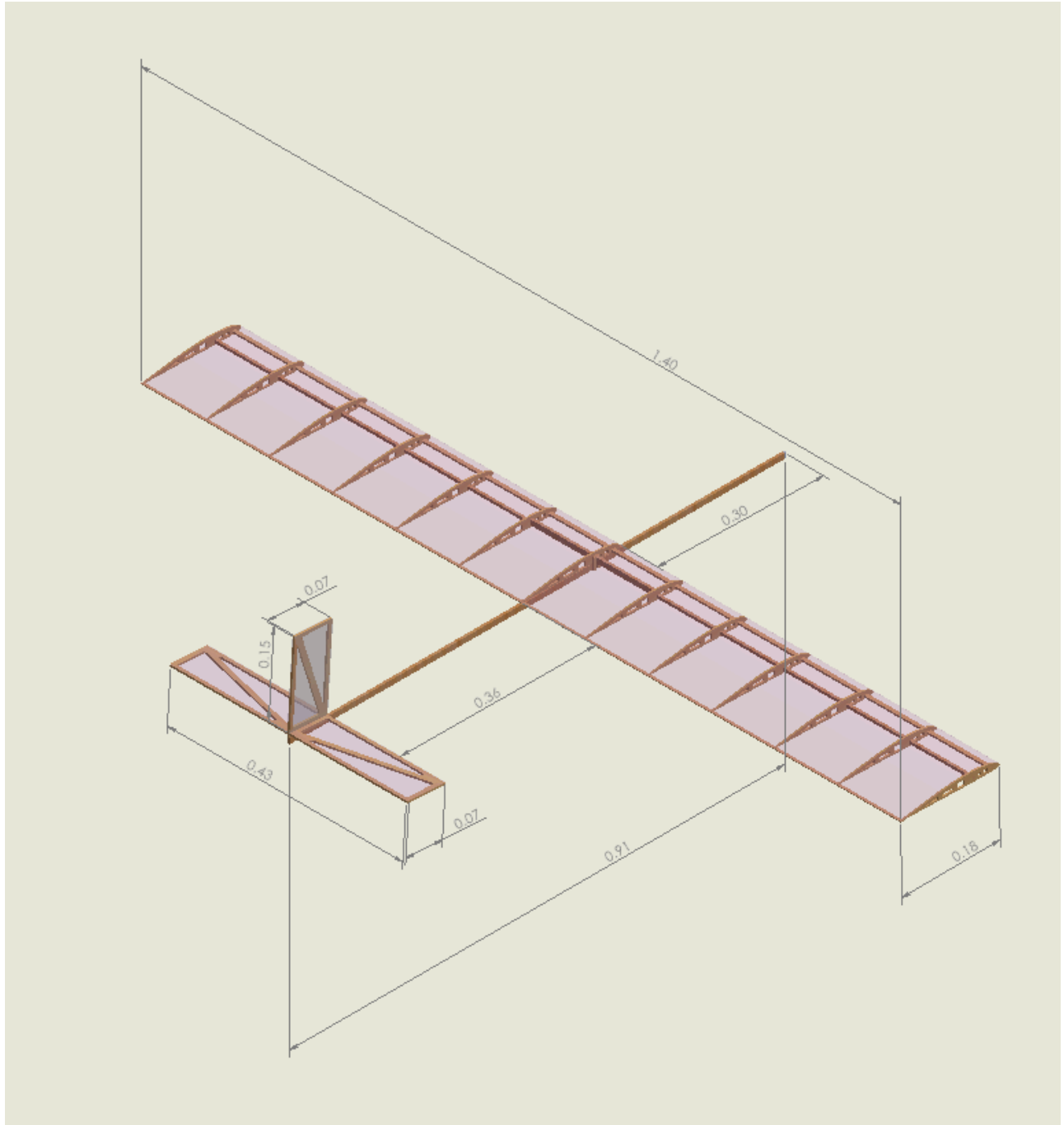


Figure 11: Isometric Drawing with Relevant Dimensions Given in Meters

Bill of Materials:

Item	Mass (g)	L(m)	Provided
Balsa $\frac{1}{8}$ inch	17.58	N/A	0.278709 m ²
Tissue Paper	11.82	N/A	N/A
Spars	40	1.8288	1.8288 m
Ballast Mass	15.6	N/A	N/A
Glue	10	N/A	50 g

Table 2: Bill of Materials

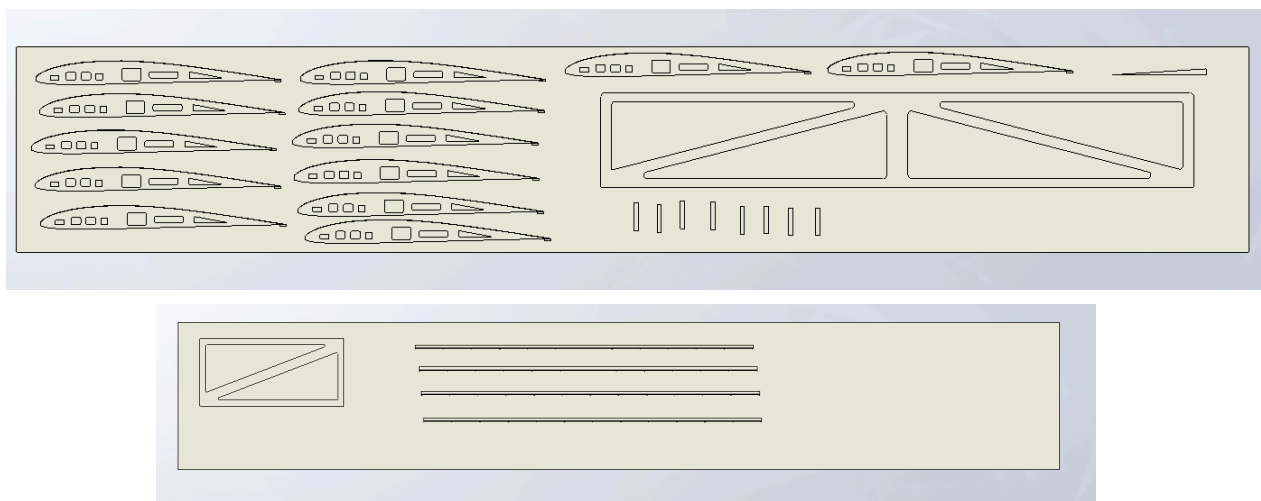


Figure 12: Drawing Package for Laser Cutting

Citations

Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 1989.

XFLR5. (n.d.). XFLR5 – Aerodynamic analysis of airfoils, wings and planes. Retrieved October 15, 2025, from <https://www.xflr5.tech/xflr5.htm>